

GANGHUA WANG

Email: wang9019@umn.edu

Website: <https://gwang.umn.edu/>

Address: 492 Ford Hall, 224 Church Street SE
Minneapolis, MN 55455

EDUCATION

University of Minnesota, Twin Cities

Ph.D. in Statistics

Aug. 2019 - present

Minneapolis, MN

Advised by [Prof. Jie Ding](#) and co-advised by [Prof. Yuhong Yang](#)

Peking University

B.S. in Statistics, Minor in Economics

Sept. 2015 - July 2019

Beijing, China

EXPERIENCE

Cisco PhD internship

Manager: Myungjin Lee, Cisco Systems Inc.

June 2022 - Aug. 2022

San Jose, CA

We studied the group fairness issue in the federated learning and proposed a bias mitigation algorithm that applies to the heterogeneous clients setting.

Cisco Research Fellow: Privacy-preserving machine learning

Supervised by Prof. Jie Ding and Prof. Yuhong Yang, University of Minnesota

Collaborated with the Cisco research team

Jan. 2021 - present

Minneapolis, MN

We formalized the definition of model privacy and proposed model defense mechanisms that provide rigorous guarantees.

Research assistant: Deep learning theory

Supervised by Prof. Jie Ding, University of Minnesota

Sept. 2021 - Present

Minneapolis, MN

We characterized the compressibility of a deep neural network using the sparsity-sensitive ℓ_q -norm ($0 < q < 1$) and proposed an adaptive pruning algorithm.

Research assistant: Data privacy

Supervised by Prof. Jie Ding, University of Minnesota

June 2020 - Aug. 2020

Minneapolis, MN

We proposed a novel framework to protect data privacy for each individual in the data collection procedure, which is different from differential privacy.

Summer internship: Component analysis in brain image

Supervised by Prof. Hongyu Zhao, Yale University

July. 2018 - Sept. 2018

New Haven, CT

We proposed to extract functional brain neural network components using the Latent Dirichlet Allocation method, which facilitates the understanding of the intrinsic structure of the human brain.

Research assistant: Community detection with co-variates

Supervised by Prof. Jinzhu Jia, Peking University

Dec. 2017 - June 2018

Beijing, China

We adopted a maximum likelihood estimation-based method to combine the information from both co-variates and the network to detect communities under the stochastic block model.

PUBLICATIONS

Journal

* indicates equal contributions

† indicates corresponding author(s)

- [1] **Ganghua Wang**, Jie Ding, and Yuhong Yang[†]. “Regression with Set-Valued Categorical Predictors”. In: *Statistica Sinica, to appear* (2022).

Submitted

- [2] **Ganghua Wang** and Jie Ding[†]. “Subset Privacy: Draw from an Obfuscated Urn”. In: *arXiv preprint* (2021).
- [3] Gen Li, **Ganghua Wang**, Yuntao Gu, and Jie Ding[†]. “Provable Identifiability of ReLU Neural Networks via LASSO Regularization”. In: *arXiv preprint* (2021).
- [4] Wenjing Yang*, **Ganghua Wang***, Enmao Diao, Vahid Tarokh, Jie Ding, and Yuhong Yang[†]. “A Theoretical Understanding of Neural Network Compression from Sparse Linear Approximation”. In: *arXiv preprint arXiv:2206.05604* (2022).

Manuscript

- [5] **Ganghua Wang**, Jie Ding, and Yuhong Yang[†]. “Model Privacy: A Framework to Understand Model Stealing Attack and Defense”. In: *arXiv preprint* (2022).
- [6] **Ganghua Wang**[†], Payani Ali, and Myungjin Lee. “Federated learning with group bias mitigation: beyond the local fairness”. In: *arXiv preprint* (2022).

HONORS AND AWARDS

Awarded by University of Minnesota, Twin Cities

Summer Research Fellowship 2020

School of Statistics First Year Scholarship 2019

Awarded by Peking University

The Academic Excellence Scholarship (3 times) 2016 - 2018

Fang Zheng Scholarship 2017

Wu Si Scholarship 2016, 2018

Freshman Scholarship 2015

TEACHING

Teaching Assistant, University of Minnesota, Twin Cities

STAT 4102 Theory of Statistics II Fall 2020

STAT 3021 Introduction to Probability and Statistics Spring 2020

STAT 3011 Introduction to Statistical Analysis Fall 2019

SERVICES

Reviewer of

The International Conference on Acoustics, Speech, & Signal Processing (ICASSP) 2022

TECHNICAL STRENGTHS

Computer Languages Python, MATLAB, R, SQL

Last edited on August 8, 2022.